

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CRYPTOGRAPHY AND NETWORK SECURITY
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Scenario-Based : 20%

Total Marks: 50

Annexure A

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

1. Secure Password Storage in Web Applications

A web application stores user passwords and needs to protect them against database breaches.

- A. Explain why storing plaintext passwords is insecure.
 - B. Suggest a secure method for storing passwords and justify your choice.
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2. Digital Signature for Document Authentication

A company wants to ensure that official documents sent electronically are authentic and unaltered.

- A. Explain how digital signatures can be used in this scenario.
 - B. Justify how this method ensures both integrity and non-repudiation.
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3. Preventing Replay Attacks in Online Transactions

An online payment system is vulnerable to replay attacks where attackers resend previously captured transactions.

- A. Explain how replay attacks occur.
 - B. Suggest a cryptographic technique to prevent such attacks and justify your answer.
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4. Secure Communication over Public Wi-Fi

A user accesses sensitive information using a public Wi-Fi network without encryption.

- A. Identify the risks involved in this scenario.
 - B. Recommend a secure communication method and justify how it protects user data.
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5. Data Integrity in Cloud Storage

An organization stores critical data in a cloud server and wants to ensure that the data is not altered.

- A. Explain how hashing can be used to verify data integrity.
- B. Suggest an additional mechanism to enhance trust and justify your answer.

1. Secure Password Storage in Web Applications

A web application stores user passwords and needs to protect them against database breaches.

A. Explain why storing plaintext passwords is insecure.

Storing passwords in plaintext is highly insecure because anyone who gains access to the database can directly read all user passwords. If attackers compromise the server, they can immediately access user accounts without performing any additional computation.

Many users reuse the same password for multiple websites, such as email, banking applications, and social media accounts. Therefore, a single database breach can compromise multiple accounts belonging to the same user.

Plaintext storage also increases the risk of insider attacks because administrators or employees with database access can view users' credentials. In addition, storing passwords in plaintext violates basic cybersecurity principles and industry standards.

Problems with plaintext password storage:

- Passwords become visible if the database is compromised.
 - Attackers can log in without any extra effort.
 - Users who reuse passwords become vulnerable on multiple platforms.
 - Insider threats increase significantly.
 - It violates security best practices.
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B. Suggest a secure method for storing passwords and justify your choice.

Passwords should be stored using secure hashing algorithms such as **bcrypt**, **Argon2**, or **scrypt**, combined with a unique salt for each user.

A hash function converts the original password into an irreversible fixed-length value. Unlike encryption, hashing is a one-way process, meaning the original password cannot be directly recovered from the hash value.

A salt is a randomly generated value added to the password before hashing. Even if two users choose the same password, the resulting hashes will be different because each user has a unique salt.

Algorithms such as bcrypt and Argon2 intentionally require more computational resources, making brute-force and dictionary attacks significantly slower.

Advantages of hashing with salt:

- Original passwords cannot be recovered easily.
 - Rainbow-table attacks are prevented.
 - Identical passwords generate different hashes.
 - Brute-force attacks become computationally expensive.
 - Security is maintained even if the database is leaked.
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2. Digital Signature for Document Authentication

A company wants to ensure that official documents sent electronically are authentic and unaltered.

A. Explain how digital signatures can be used in this scenario.

Digital signatures use public-key cryptography to verify the authenticity of electronic documents.

The sender first generates a hash of the document using a hashing algorithm such as SHA-256. This hash value is then encrypted using the sender's private key to create the digital signature.

The document and its signature are transmitted to the receiver.

Upon receiving the document, the receiver calculates the hash of the received document and decrypts the digital signature using the sender's public key. The receiver then compares both hash values.

If the values match, the document is authentic and has not been modified during transmission.

Steps involved:

1. Generate the document hash.
 2. Encrypt the hash using the sender's private key.
 3. Attach the digital signature to the document.
 4. Send the document to the receiver.
 5. Decrypt the signature using the public key.
 6. Compare both hash values.
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B. Justify how this method ensures both integrity and non-repudiation.

Integrity

Data integrity means that the information remains unchanged during transmission.

If even a single character in the document is modified, the hash value changes completely. Since the receiver compares hash values, any alteration can be detected immediately.

Therefore, digital signatures guarantee data integrity.

Non-repudiation

Non-repudiation means that the sender cannot deny sending the document.

Since only the sender possesses the private key used to generate the signature, the signature acts as proof that the document originated from the sender.

Thus, digital signatures provide legal and technical evidence of ownership.

Benefits of digital signatures:

- Ensure document authenticity.
 - Detect unauthorized modifications.
 - Prevent denial by the sender.
 - Provide legal validity.
 - Improve trust in electronic communication.
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3. Preventing Replay Attacks in Online Transactions

An online payment system is vulnerable to replay attacks in which attackers resend previously captured transactions.

A. Explain how replay attacks occur.

A replay attack occurs when an attacker intercepts a valid communication between two parties and later retransmits the same message.

For example, suppose a customer sends a payment request to transfer ₹5000. If an attacker captures this request and sends it again, the system may process the same transaction multiple times.

Replay attacks exploit systems that fail to verify whether a message has already been used.

Steps in a replay attack:

1. The attacker intercepts a valid message.
2. The attacker stores the message.
3. The attacker retransmits the message later.
4. The server accepts the message as valid.
5. Unauthorized actions occur.

Consequences:

- Duplicate transactions.
 - Financial losses.
 - Unauthorized access.
 - Loss of customer trust.
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B. Suggest a cryptographic technique to prevent such attacks and justify your answer.

Replay attacks can be prevented using the following techniques:

1. Nonce

A nonce is a randomly generated number used only once in a transaction.

The server checks whether the nonce has already been used. If the same nonce appears again, the request is rejected.

2. Timestamp

A timestamp records the exact time at which the transaction was created.

Requests older than a specified time limit are rejected.

3. Session Tokens

Session tokens uniquely identify each communication session. Once the session ends, the token becomes invalid.

Justification:

- Nonces guarantee uniqueness.
- Timestamps prevent reuse of old messages.
- Session tokens expire automatically.

- Duplicate requests are detected and rejected.

These mechanisms ensure that each transaction is processed only once.

4. Secure Communication over Public Wi-Fi

A user accesses sensitive information using a public Wi-Fi network without encryption.

A. Identify the risks involved in this scenario.

Public Wi-Fi networks are often unsecured and can expose users to several cybersecurity threats.

Risks involved:

1. Eavesdropping

Attackers can intercept data packets and read sensitive information such as usernames and passwords.

2. Man-in-the-Middle Attack

Attackers position themselves between the user and the server and secretly modify or monitor communications.

3. Session Hijacking

Hackers steal session cookies and impersonate legitimate users.

4. Data Manipulation

Attackers may alter messages transmitted over the network.

5. Identity Theft

Personal and financial information may be stolen.

Consequences:

- Leakage of confidential information.
- Financial loss.
- Unauthorized account access.
- Privacy violations.
- Data corruption.

B. Recommend a secure communication method and justify how it protects user data.

The recommended solutions are **HTTPS (TLS/SSL)** and **Virtual Private Networks (VPNs)**.

HTTPS (TLS/SSL)

Transport Layer Security encrypts data transmitted between the client and the server.

Advantages:

- Protects confidentiality.
- Prevents eavesdropping.
- Verifies server identity.
- Detects tampering.

VPN

A VPN creates an encrypted tunnel between the user's device and the destination server.

Advantages:

- Hides internet traffic.
- Protects data on public networks.
- Prevents packet sniffing.
- Improves privacy.

Using HTTPS and VPN together provides strong protection against attacks on public Wi-Fi networks.

5. Data Integrity in Cloud Storage

An organization stores critical data in a cloud server and wants to ensure that the data is not altered.

A. Explain how hashing can be used to verify data integrity.

Hashing is a process that converts data into a fixed-length value called a hash or message digest.

Before uploading a file to the cloud, the organization computes a hash value using algorithms such as SHA-256.

After downloading the file, the organization computes the hash again and compares it with the original value.

Verification process

1. Generate the hash before uploading.
2. Store the hash securely.
3. Download the file from the cloud.
4. Generate a new hash.
5. Compare both hash values.

If both hashes match, the file remains unchanged. If they differ, the file has been modified.

Benefits of hashing

- Fast verification.
 - Detects even minor modifications.
 - Ensures data integrity.
 - Requires minimal storage space.
 - Easy to implement.
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B. Suggest an additional mechanism to enhance trust and justify your answer.

To improve trust, organizations can use **digital signatures** or **Message Authentication Codes (MACs)**.

Digital Signatures

Digital signatures use private and public keys to verify both authenticity and integrity.

Advantages:

- Verify the source of data.
- Detect tampering.
- Provide non-repudiation.
- Increase trust.

Message Authentication Codes (MACs)

A MAC combines a secret key with the message to generate an authentication code.

Advantages:

- Ensures data integrity.
- Verifies authenticity.
- Prevents unauthorized modifications.
- Efficient and secure.

Hashing alone only verifies whether data has changed. Digital signatures and MACs additionally verify who created the data, making them more trustworthy.

Conclusion

Modern cybersecurity relies heavily on cryptographic techniques such as hashing, digital signatures, encryption, nonces, timestamps, TLS, VPNs, and MACs. These techniques ensure confidentiality, integrity, authenticity, and non-repudiation.

Proper implementation of these mechanisms protects systems against attacks such as password theft, replay attacks, data tampering, and unauthorized access.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand